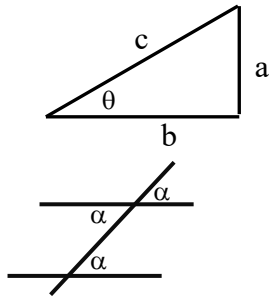
PHYSICS 1101.200 Group Problem 6

This is a closed book and closed notes group problem with no connection to any source of information. Calculators are permitted only to do mathematical calculations. No devices that can contact the internet are allowed. The **ONLY** physics formulas that may be used are those given below. Define all symbols and justify all mathematical expressions used. Make sure to state all assumptions & approximations used to solve a problem. If this were a quiz, partial credit would be given for a well-communicated problem-solving framework based on correct physics. Each problem is worth 25 points: **In the context of a unified solution**, credit will be awarded as follows: a useful sketch that shows all relevant quantities, listing the known quantities, and writing the quantity to be determined is 7 points; writing the physics principles useful to solve the problem, writing any assumptions or approximations needed, drawing useful physics diagrams, and writing the equations that apply the physics principles in terms of the quantities from your sketch is 6 points; planning the solution by describing how to use those physics equations to construct an algebraic answer is 5 points; solving those equations for a symbolic result is 3 points; checking the units & reasonableness of the symbolic result is 2 points; calculating a numerical result & stating why its units are correct and giving a reason why that result is not unreasonable is 2 points. If no numerical answer is required, 1 point will be added the planning the solution and 1 point to solving the equations and checking the units & reasonableness of the solution.

Useful Mathematical Relationships:



For a right triangle: $\sin\theta = \frac{a}{c}$, $\cos\theta = \frac{b}{c}$, $\tan\theta = \frac{a}{b}$,

$a^2 + b^2 = c^2$, $\sin^2\theta + \cos^2\theta = 1$

For a circle: $C = 2\pi R$, $A = \pi R^2$

$V = \frac{4}{3}\pi R^3$, $S = 4\pi R^2$

If $Ax^2 + Bx + C = 0$, $x = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, $(a+b)^2 = a^2 + 2ab + b^2$

Fundamental Concepts, Principles, and Definitions:

$\sum \vec{F} = m\vec{a}$	$\vec{F}_{A\text{ on }B} = -\vec{F}_{B\text{ on }A}$	$a_{x\,ave} = \frac{\Delta v_x}{\Delta t}$	$v_{x\,ave} = \frac{\Delta x}{\Delta t}$	$\tau = rF_{\perp}$	$KE = \frac{1}{2}mv^2$

$E_{final} - E_{initial} = E_{in} - E_{out}$

Under Certain Conditions:

$F = mg$	$F = \mu N$	$F_{max} = \mu N$	$F = k\Delta L$	$\sum \vec{F} = 0$	$\sum \vec{\tau} = 0$

$E_{transfer} = F_{\parallel} d$

Useful constants: 1 mile = 5280 ft, g = 9.8 m/s^2 = 32 ft/s^2

You are investigating the use of rescue sleds pulled by snowmobiles for snow emergencies. To determine how this could work, you first need to measure the coefficient of friction of a typical snowpack on a moving fully loaded 160 kg sled. To accomplish this, you take a fully loaded sled up a hill, which is at 15° to the horizontal and covered with a typical snowpack. You then release the sled so that it slides down the hill. Borrowing a police radar gun, you determine that the sled goes 10 miles/hour and is speeding up when it is 120 feet from its release point. Review your problem-solving process by discussing within your group and writing down every decision you make to solve this problem. Make sure everyone in the group understands why the decisions are needed and what alternative decisions are also valid that might result in another path to a solution.